

Monte Carlo Fisher Information for Generalized Linear Mixed Model

Julius Olaifa

October 9, 2026

1 Parameter Setup and Likelihood

Let

$$\Psi = (\beta, \Gamma, \theta)$$

denote the full parameter vector, where β contains the fixed-effect parameters, Γ contains the variance-covariance parameters associated with the random effects, and θ denotes the dispersion parameter.

The log-likelihood is written as

$$\ell(\Psi) = \log L(\Psi),$$

where $L(\Psi)$ denotes the likelihood function.

For n independent sampling units, the log-likelihood can be expressed as

$$\ell(\Psi) = \sum_{i=1}^n \ell_i(\Psi),$$

with corresponding score contribution

$$U_i(\Psi) = \nabla_{\Psi} \ell_i(\Psi).$$

The total score is therefore

$$U(\Psi) = \sum_{i=1}^n U_i(\Psi).$$

The Hessian matrix of the full log-likelihood is

$$H(\Psi) = \nabla_{\Psi}^2 \ell(\Psi).$$

Under standard regularity conditions, the Fisher information matrix for the full sample is

$$I_n(\Psi) = -\mathbb{E}_{\Psi}[H(\Psi)] = \mathbb{E}_{\Psi}[U(\Psi)U(\Psi)^{\top}].$$

Since the sampling units are independent, information is additive, so that

$$I_n(\Psi) = \sum_{i=1}^n I_i(\Psi).$$

If the sampling units are identically distributed, then

$$\boxed{I_n(\Psi) = nI_1(\Psi),}$$

where

$$I_1(\Psi) = \mathbb{E}_\Psi[U_i(\Psi)U_i(\Psi)^\top].$$

1.1 Monte Carlo Approximation

Under the standard regularity condition that the score has mean zero,

$$\mathbb{E}_\Psi[U_i(\Psi)] = 0,$$

the per-unit Fisher information can equivalently be written as

$$I_1(\Psi) = \text{Var}_\Psi(U_i(\Psi)) = \mathbb{E}_\Psi[U_i(\Psi)U_i(\Psi)^\top].$$

The expectation defining the Fisher information can be approximated using Monte Carlo simulation. Suppose R independent replicates are generated from the model under the parameter value Ψ , and let

$$U_i^{(r)}(\Psi), \quad r = 1, \dots, R,$$

denote the score contribution from one sampling unit in the r th Monte Carlo replicate.

Although the score has theoretical mean zero, its empirical mean over a finite number of Monte Carlo replicates will generally not be exactly zero. Define the Monte Carlo mean score as

$$\bar{U}_i(\Psi) = \frac{1}{R} \sum_{r=1}^R U_i^{(r)}(\Psi).$$

The per-unit Fisher information is therefore estimated using the centered score vectors,

$$\hat{I}_1(\Psi) = \frac{1}{R} \sum_{r=1}^R \left(U_i^{(r)}(\Psi) - \bar{U}_i(\Psi) \right) \left(U_i^{(r)}(\Psi) - \bar{U}_i(\Psi) \right)^\top.$$

Consequently, for n independent and identically distributed sampling units,

$$\boxed{\hat{I}_n(\Psi) = n\hat{I}_1(\Psi) = \frac{n}{R} \sum_{r=1}^R \left(U_i^{(r)}(\Psi) - \bar{U}_i(\Psi) \right) \left(U_i^{(r)}(\Psi) - \bar{U}_i(\Psi) \right)^\top.}$$

1.2 Fisher Information of the Variance Components

Let γ denote the vector containing the variance-covariance parameters associated with the random effects. The Fisher information relevant to γ , after accounting for the remaining parameters in Ψ , is obtained from the $\gamma\gamma$ block of the inverse of the full Fisher information matrix. Specifically,

$$I_{\gamma\gamma}^{\text{eff}} = \left[\{I_n(\Psi)^{-1}\}_{\gamma\gamma} \right]^{-1}.$$

That is, the full Fisher information matrix $I_n(\Psi)$ is first inverted, the block corresponding to the variance-covariance parameters γ is extracted, and this block is then inverted to obtain the effective Fisher information for γ .

2 Derivation of the Score Vector

2.1 GLMM

Consider the generalized linear mixed model

$$m(\mu_{ij}) = X_{ij}^\top \beta + Z_{ij}^\top b_i,$$

where

$$b_i \sim N(0, \Gamma).$$

Define the linear predictor

$$\eta_{ij} = X_{ij}^\top \beta + Z_{ij}^\top b_i,$$

so that

$$m(\mu_{ij}) = \eta_{ij}, \quad \mu_{ij} = m^{-1}(\eta_{ij}).$$

Conditional on the random effect b_i , assume that

$$y_{ij} \mid b_i \sim f(\mu_{ij}, \theta),$$

or, equivalently,

$$y_{ij} \mid b_i \sim f(m^{-1}(\eta_{ij}), \theta).$$

2.2 Conditional Log Likelihood

For a single observation, let

$$\ell_{ij} = \log f(y_{ij} \mid b_i),$$

and define

$$g_{ij} = \frac{\partial \ell_{ij}}{\partial \eta_{ij}}.$$

Conditional on b_i , the observations within sampling unit i are assumed to be independent. The conditional likelihood contribution from unit i is therefore

$$L_i(b_i) = \prod_j f(y_{ij} | b_i).$$

The corresponding conditional log-likelihood is

$$\ell_i(b_i) = \log L_i(b_i) = \sum_j \log f(y_{ij} | b_i).$$

Define the score of the conditional log-likelihood with respect to the random effect b_i as

$$S_i = \frac{\partial \ell_i(b_i)}{\partial b_i},$$

and the corresponding Hessian as

$$H_i = \frac{\partial^2 \ell_i(b_i)}{\partial b_i \partial b_i^\top}.$$

Thus, if $b_i \in \mathbb{R}^q$, then

$$S_i \in \mathbb{R}^q, \quad H_i \in \mathbb{R}^{q \times q}.$$

$$\frac{\partial \eta_{ij}}{\partial \beta} = X_{ij}, \quad \frac{\partial \eta_{ij}}{\partial b_i} = Z_{ij}.$$

Using the chain rule, the score with respect to the fixed-effect parameters is

$$U_{\beta,i} = \frac{\partial \ell_i}{\partial \beta} = \sum_j \frac{\partial \ell_{ij}}{\partial \eta_{ij}} \frac{\partial \eta_{ij}}{\partial \beta} = \sum_j g_{ij} X_{ij} = X_i^\top g_i.$$

Similarly, the score with respect to the random effect is

$$U_{b,i} = \frac{\partial \ell_i}{\partial b_i} = \sum_j \frac{\partial \ell_{ij}}{\partial \eta_{ij}} \frac{\partial \eta_{ij}}{\partial b_i} = \sum_j g_{ij} Z_{ij} = Z_i^\top g_i.$$

Note that

$$S_i = U_{b,i} = \frac{\partial \ell_i}{\partial b_i}$$

We have obtained the score with respect to the fixed-effect parameters β and the random effects b_i . However, b_i is not itself part of the parameter vector Ψ . Rather,

$$b_i \sim N(0, \Gamma),$$

and the parameter of interest is the variance-covariance matrix Γ .

To obtain the score with respect to Γ , we first derive the first and second derivatives of the conditional likelihood with respect to b_i .

2.3 Conditional Likelihood

Since

$$\ell_i(b_i) = \log L_i(b_i),$$

the conditional likelihood can equivalently be written as

$$L_i(b_i) = \exp\{\ell_i(b_i)\}.$$

Differentiating the conditional likelihood with respect to b_i gives

$$\frac{\partial L_i(b_i)}{\partial b_i} = \exp\{\ell_i(b_i)\} \frac{\partial \ell_i(b_i)}{\partial b_i}.$$

Using the definition of S_i ,

$$\boxed{\frac{\partial L_i(b_i)}{\partial b_i} = L_i(b_i) S_i.}$$

Differentiating once more with respect to $b_i^\top$,

$$\frac{\partial^2 L_i(b_i)}{\partial b_i \partial b_i^\top} = \frac{\partial}{\partial b_i^\top} \{L_i(b_i) S_i\}.$$

By the product rule,

$$\frac{\partial^2 L_i(b_i)}{\partial b_i \partial b_i^\top} = S_i \left(\frac{\partial L_i(b_i)}{\partial b_i} \right)^\top + L_i(b_i) \frac{\partial S_i}{\partial b_i^\top}.$$

Since

$$\frac{\partial L_i(b_i)}{\partial b_i} = L_i(b_i) S_i$$

and

$$\frac{\partial S_i}{\partial b_i^\top} = H_i,$$

it follows that

$$\boxed{\frac{\partial^2 L_i(b_i)}{\partial b_i \partial b_i^\top} = L_i(b_i) (H_i + S_i S_i^\top).}$$

For convenience, define

$$\boxed{A_i = H_i + S_i S_i^\top.}$$

Hence,

$$\boxed{\frac{\partial^2 L_i(b_i)}{\partial b_i \partial b_i^\top} = L_i(b_i) A_i.}$$

2.4 Averaging over $\mathbf{b}$

The likelihood $L_i(b_i)$ is conditional on a particular value of b_i . To obtain the likelihood of the observed data, the unobserved random effect must therefore be integrated out with respect to its distribution. The marginal likelihood contribution of sampling unit i is

$$L_i(\beta, \Gamma, \theta) = \int L_i(b_i) \phi(b_i; 0, \Gamma) db_i,$$

where $\phi(b_i; 0, \Gamma)$ denotes the multivariate normal density with mean zero and covariance matrix Γ .

Equivalently, the marginal likelihood can be written as an expectation over the distribution of the random effect,

$$L_i(\beta, \Gamma, \theta) = \mathbb{E}_{b_i \sim N(0, \Gamma)} [L_i(b_i)].$$

Thus, Γ enters the marginal likelihood through the distribution over which the conditional likelihood $L_i(b_i)$ is averaged. This provides the connection between the derivatives of $L_i(b_i)$ with respect to b_i derived above and the score with respect to Γ .

2.4.1 Taylor Approximation

To study the marginal likelihood near $\Gamma = 0$, we consider a second-order Taylor expansion of the conditional likelihood $L_i(b_i)$ around $b_i = 0$:

$$L_i(b_i) \approx L_i(0) + \left. \frac{\partial L_i(b_i)}{\partial b_i} \right|_{b_i=0}^\top b_i + \frac{1}{2} b_i^\top \left. \frac{\partial^2 L_i(b_i)}{\partial b_i \partial b_i^\top} \right|_{b_i=0} b_i.$$

From the previous results,

$$\frac{\partial L_i(b_i)}{\partial b_i} = L_i(b_i) S_i$$

and

$$\frac{\partial^2 L_i(b_i)}{\partial b_i \partial b_i^\top} = L_i(b_i) A_i.$$

Evaluating these expressions at $b_i = 0$, the Taylor expansion becomes

$$L_i(b_i) \approx L_i(0) + L_i(0) S_i^\top b_i + \frac{1}{2} L_i(0) b_i^\top A_i b_i,$$

where S_i and A_i are evaluated at $b_i = 0$.

Factoring out $L_i(0)$,

$$L_i(b_i) \approx L_i(0) \left[1 + S_i^\top b_i + \frac{1}{2} b_i^\top A_i b_i \right].$$

Taking expectation with respect to $b_i \sim N(0, \Gamma)$ gives

$$\mathbb{E}_{b_i}[L_i(b_i)] \approx L_i(0) \left[1 + \mathbb{E}_{b_i}(S_i^\top b_i) + \frac{1}{2} \mathbb{E}_{b_i}(b_i^\top A_i b_i) \right].$$

Since

$$\mathbb{E}_{b_i}[b_i] = 0,$$

the linear term vanishes:

$$\mathbb{E}_{b_i}(S_i^\top b_i) = S_i^\top \mathbb{E}_{b_i}[b_i] = 0.$$

For the quadratic term, using

$$b_i^\top A_i b_i = \text{tr}(A_i b_i b_i^\top),$$

we obtain

$$\mathbb{E}_{b_i}(b_i^\top A_i b_i) = \text{tr}(A_i \mathbb{E}_{b_i}[b_i b_i^\top]).$$

Because

$$b_i \sim N(0, \Gamma),$$

we have

$$\mathbb{E}_{b_i}[b_i b_i^\top] = \Gamma.$$

Therefore,

$$\mathbb{E}_{b_i}(b_i^\top A_i b_i) = \text{tr}(A_i \Gamma).$$

Recalling that the marginal likelihood is

$$L_i(\beta, \Gamma, \theta) = \mathbb{E}_{b_i}[L_i(b_i)],$$

we obtain the approximation

$$\boxed{L_i(\beta, \Gamma, \theta) \approx L_i(0) \left[1 + \frac{1}{2} \text{tr}(A_i \Gamma) \right].}$$

2.5 Score of the Variance Components

From the approximation to the marginal likelihood,

$$L_i(\beta, \Gamma, \theta) \approx L_i(0) \left[1 + \frac{1}{2} \text{tr}(A_i \Gamma) \right],$$

the corresponding marginal log-likelihood is

$$\ell_i(\beta, \Gamma, \theta) \approx \log \left\{ L_i(0) \left[1 + \frac{1}{2} \text{tr}(A_i \Gamma) \right] \right\}.$$

Therefore,

$$\ell_i(\beta, \Gamma, \theta) \approx \ell_i(0) + \log \left[1 + \frac{1}{2} \text{tr}(A_i \Gamma) \right].$$

Using the first-order approximation

$$\log(1+x) \approx x, \quad x \approx 0,$$

and noting that

$$\text{tr}(A_i \Gamma) \rightarrow 0 \quad \text{as} \quad \Gamma \rightarrow 0,$$

we obtain

$$\boxed{\ell_i(\beta, \Gamma, \theta) \approx \ell_i(0) + \frac{1}{2} \text{tr}(A_i \Gamma).}$$

The corresponding differential of the log-likelihood is

$$d\ell_i = \frac{1}{2} \text{tr}(A_i d\Gamma).$$

Using the matrix inner product

$$\langle B, C \rangle = \text{tr}(B^\top C),$$

and noting that A_i is symmetric, this can be written as

$$d\ell_i = \left\langle \frac{1}{2} A_i, d\Gamma \right\rangle.$$

Hence, as a gradient on the space of symmetric matrices,

$$\boxed{\nabla_\Gamma \ell_i|_{\Gamma=0} = \frac{1}{2} A_i.}$$

This matrix expression represents the score with respect to the covariance matrix Γ . Since Γ is symmetric, its distinct entries may subsequently be collected into a vector of variance-component parameters when constructing the full score vector.

To obtain the score with respect to the distinct variance-covariance parameters, let

$$\Gamma = \begin{pmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{12} & \gamma_{22} \end{pmatrix}$$

and

$$A_i = \begin{pmatrix} A_{11,i} & A_{12,i} \\ A_{12,i} & A_{22,i} \end{pmatrix}.$$

Since both A_i and Γ are symmetric,

$$\text{tr}(A_i \Gamma) = A_{11,i} \gamma_{11} + 2A_{12,i} \gamma_{12} + A_{22,i} \gamma_{22}.$$

Therefore,

$$\ell_i(\beta, \Gamma, \theta) \approx \ell_i(0) + \frac{1}{2} A_{11,i} \gamma_{11} + A_{12,i} \gamma_{12} + \frac{1}{2} A_{22,i} \gamma_{22}.$$

Differentiating with respect to the three distinct variance-covariance parameters gives

$$\left. \frac{\partial \ell_i}{\partial \gamma_{11}} \right|_{\Gamma=0} = \frac{1}{2} A_{11,i},$$

$$\left. \frac{\partial \ell_i}{\partial \gamma_{12}} \right|_{\Gamma=0} = A_{12,i},$$

and

$$\left. \frac{\partial \ell_i}{\partial \gamma_{22}} \right|_{\Gamma=0} = \frac{1}{2} A_{22,i}.$$

Hence, defining

$$\gamma = \begin{pmatrix} \gamma_{11} \\ \gamma_{12} \\ \gamma_{22} \end{pmatrix},$$

the score for the variance components is

$$U_{\gamma,i} = \left. \frac{\partial \ell_i}{\partial \gamma} \right|_{\Gamma=0} = \begin{pmatrix} \frac{1}{2} A_{11,i} \\ A_{12,i} \\ \frac{1}{2} A_{22,i} \end{pmatrix}.$$

3 Distribution Specific

In general, the first derivative of the individual log-likelihood with respect to the linear predictor can be written using the chain rule as

$$g_{ij} = \frac{\partial \ell_{ij}}{\partial \eta_{ij}} = \frac{\partial \ell_{ij}}{\partial \mu_{ij}} \frac{\partial \mu_{ij}}{\partial \eta_{ij}}.$$

Similarly, define

$$h_{ij} = \frac{\partial^2 \ell_{ij}}{\partial \eta_{ij}^2}.$$

Applying the chain rule gives

$$h_{ij} = \frac{\partial^2 \ell_{ij}}{\partial \mu_{ij}^2} \left(\frac{\partial \mu_{ij}}{\partial \eta_{ij}} \right)^2 + \frac{\partial \ell_{ij}}{\partial \mu_{ij}} \frac{\partial^2 \mu_{ij}}{\partial \eta_{ij}^2}.$$

Thus, the explicit forms of g_{ij} and h_{ij} depend on both the conditional distribution of y_{ij} and the choice of link function.